

Complete Midterm Roundup

Foundations, The Stack, Protocols, and Command Line

Instructor: Bernal Manzanilla

Concordia University of Edmonton

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Week 1: The Telegraph & Physical Limits

Before we had the modern internet, we had the Transatlantic Telegraph. It taught us the physical limits of communication.

The Problem of Dispersion

When you send an electrical pulse (a dot or a dash) down a massive copper wire, the energy "spreads out" or smears as it travels.

Inter-Symbol Interference (ISI)

- If you send pulses too fast, the tail of the first pulse bleeds into the beginning of the second pulse.
- The receiver just sees a continuous, unreadable blur.
- **Conclusion:** There is a hard physical limit to the **Inter-Symbol Rate** (Baud Rate) of any cable.

Week 1: Sharing the Wire (Multiplexing)

How do millions of people share the same cables without their signals crashing into each other?

1. FDM (Frequency Division Multiplexing)

- **How it works:** The cable is divided into different frequency "lanes."
- **The Analogy:** Like FM Radio. 91.7 FM and 100.3 FM exist in the air at the exact same time, but they don't interfere.
- **Result:** You get your own frequency 100% of the time.

2. TDM (Time Division Multiplexing)

- **How it works:** Users take turns.
- **The Analogy:** A panel discussion with one microphone. Everyone gets to use the entire microphone, but only for 30 seconds at a time.

Week 2: Circuit vs. Packet Switching

Circuit Switching (The Old Telephone Network)

- Uses FDM or TDM to reserve a dedicated, physical path just for you.
- **The Flaw:** If you are on the phone but not speaking, the wire is completely wasted. No one else can use your reserved slot.

Packet Switching (The Modern Internet)

- Data is broken into small chunks called "Packets."
- Everyone shares the wire equally. There are no reservations.
- **The Queue:** If you and I try to send a packet at the exact same millisecond, the router puts one of them in a "buffer" (a waiting line) until the wire is clear.

The Protocol Stack: The Russian Doll

Data never travels naked. It uses **Encapsulation**.

The Russian Doll Analogy:

- Your email (Data) is put inside a Transport box (TCP).
- The TCP box is put inside a Network box (IP).
- The IP box is put inside a Link box (Ethernet).

To "parse" a packet, the receiving computer simply peels off the layers one by one, reading the headers.

Memorize: The Packages (PDUs)

Every layer calls its specific "box" by a different name. You must know these for the midterm!

Layer Name	Package Name (PDU)
Layer 7 (Application)	MESSAGE
Layer 4 (Transport)	SEGMENT
Layer 3 (Network)	DATAGRAM
Layer 2 (Link)	FRAME
Layer 1 (Physical)	BITS (0s and 1s)

Where are we today?

Layer 7: Application (TODAY'S TOPIC) *(Data Unit: Message)*

Protocols: **HTTP (Web)**, SMTP (Mail), DNS

Content: User Data (HTML, Images, Text)

Layer 4: Transport *(Data Unit: Segment)*

Protocols: TCP, UDP

Header: Source/Dest **Ports**, Sequence Num, Checksum

Layer 3: Network *(Data Unit: Datagram)*

Protocols: IP (IPv4/IPv6), ICMP

Header: Source/Dest **IP Addresses**, TTL

Layer 2: Link / Physical *(Data Unit: Frame)*

Protocols: Ethernet, Wi-Fi, ARP

Header: Source/Dest **MAC Addresses**, Error Check

Source: Kurose & Ross (9th Ed): Figure 1.24 (Protocol Stack)

What is HTTP?

Definition

Hyper**T**ext **T**ransfer **P**rotocol is the language web browsers and servers use to talk to each other.

How it works (The Conversation):

- **Client (Chrome):** "I want to see /index.html."
- **Server (Apache):** "Here is the file." (or "I can't find it")

Key Concept: Stateless

- The server forgets you the moment it sends the file.
- It does NOT remember if you visited 5 seconds ago. (That's why we need Cookies!)

Source: Kurose & Ross: Section 2.2.1 (Overview of HTTP)

Request Method 1: GET

GET = "Give me something"

Used when you want to **retrieve** data (load a page, view an image).

Characteristics:

- Data is visible in the URL (e.g., search?q=cat).
- **NOT** secure for passwords.
- Can be bookmarked.

```
GET /index.html HTTP/1.1
Host: www.concordia.ab.ca
User-Agent: Mozilla/5.0 (Windows 10)
```

Request Method 2: POST

POST = "Take this"

Used when you want to **send** data to the server (Login, Upload, Form Submit).

Characteristics:

- Data is hidden inside the "Body" of the message.
- **Secure** (if using HTTPS).
- Cannot be bookmarked.

```
POST /login.php HTTP/1.1
Host: www.concordia.ab.ca

username=student&password=12345
```

The Server's Reply: Status Codes

When the server answers, it gives a 3-digit code to say what happened.

Range	Meaning	Famous Examples
200-299	Success	200 OK (Here is your file)
300-399	Moved	301 Moved Permanently (Go here instead)
400-499	User Error	404 Not Found (You typed it wrong)
		403 Forbidden (No permission)
500-599	Server Error	500 Internal Error (My code crashed)

Source: Kurose & Ross: Section 2.2.3 (HTTP Message Format)

Layer 7: DNS (The Phonebook)

Computers need IP addresses to communicate, but humans prefer names. **DNS (Domain Name System)** translates names into numbers.

The Three Core Records:

- ❶ **A Record (Address):** The most common. Maps a hostname directly to an IPv4 address (e.g., `google.com` → `8.8.8.8`).
- ❷ **CNAME Record (Canonical Name):** An alias. Maps a nickname to a real name (e.g., `www.google.com` → `google.com`).
- ❸ **MX Record (Mail Exchange):** Tells the internet where to deliver emails for that domain.

Layer 4: Understanding Ports

We know the IP address gets the data to the correct *building*. But which application gets the data? Spotify? Chrome? Discord?

The Apartment Building Analogy

- **IP Address:** The street address of the entire building.
- **Port Number:** The specific apartment number inside.

The "Celebrity" Ports (Memorize These!):

- **Port 80:** HTTP (Unencrypted Web)
- **Port 443:** HTTPS (Secure Web)
- **Port 53:** DNS (The Phonebook)
- **Port 22:** SSH (Secure Terminal Admin)

Layer 4: UDP vs. TCP

UDP (The Fast Courier)

- **Header:** 8 Bytes (Tiny).
- **Style:** "Fire and Forget."
- **Reliability:** None. It does not care if packets are lost.
- **Uses:** Streaming video, DNS queries.

TCP (The Big Brother)

- **Header:** 20 Bytes (A monster contract).
- **Style:** Requires a Handshake.
- **Reliability:** Guaranteed. It retransmits lost data.
- **Uses:** Web (HTTP), Email, File downloads.

Deep Dive: How TCP Works

1. Sequence Numbers (Counting Bytes)

- TCP does *not* count packets (1, 2, 3).
- It counts **BYTES**.
- If packet #1 holds bytes 0-1000, the Sequence Number for packet #2 is **1001**. This allows perfect error correction.

2. Window Size (The Traffic Light)

- This is TCP's **Flow Control**.
- If a massive server sends data to a slow phone, the phone's buffer will overflow.
- The phone tells the server: "My Window Size is 5000 bytes."
- Translation: "I only have room for 5000 bytes right now. Send more than that, and you will crash me."

TCP: Handshakes & Flags

Inside the TCP header, there are 1-bit switches called **Flags**.

The 3-Way Handshake:

- 1 **[SYN]**: Client says "Hello, I want to connect."
- 2 **[SYN, ACK]**: Server says "I hear you, and I also want to connect."
- 3 **[ACK]**: Client says "Great, let's start sending data."

Other Critical Flags:

- **[F] FIN (Finish)**: A polite goodbye to end the connection.
- **[R] RST (Reset)**: The Emergency Brake! "Error, stop talking to me immediately!"
- **[P] PSH (Push)**: The Impatient Waiter. "Don't wait for the buffer to fill up; send this data to Chrome immediately!"

Layer 3: ICMP and Traceroute

ICMP (Internet Control Message Protocol)

- This is strictly a diagnostic tool.
- It is used to send "Echo Requests" (Ping) and error messages. It never carries user data.

TTL (Time to Live)

- A self-destruct timer on an IP Datagram. Every time a router touches the packet, the TTL drops by 1. If it hits 0, the packet is destroyed to prevent infinite loops.

Traceroute (The Clever Hack)

- Discovers the path by sending packets with short TTLs (1, then 2, then 3...).
- It forces routers along the path to send back "Time Exceeded" ICMP errors, revealing their IP addresses.

The Command Line Toolkit

You must know which tool to use for a specific scenario!

- `nslookup`: Query the DNS phonebook to find an IP.
- `ping`: Send an ICMP Echo to verify a server is alive.
- `tracert`: Discover the routers (hops) along the path.
- `curl -o file.txt`: Act like a web browser. The `-o` saves the file and displays the transfer statistics table.

Packet Inspection (tcpdump & grep)

Network admins use the terminal to parse massive amounts of data.

1. Capture the Traffic

```
sudo tcpdump -nv host gaia.cs.umass.edu > log.txt
```

The > redirects the output to a text file instead of the screen.

2. Filter with Grep (Regex)

```
grep "Flags \[S\]" log.txt
```

Because brackets [] are special code in Regular Expressions, we must use a backslash \ to "escape" them. This tells the computer to look for the literal bracket character to find the SYN flag.

Reading the Output (Checksums)

When you filter the output, you will see something like this:

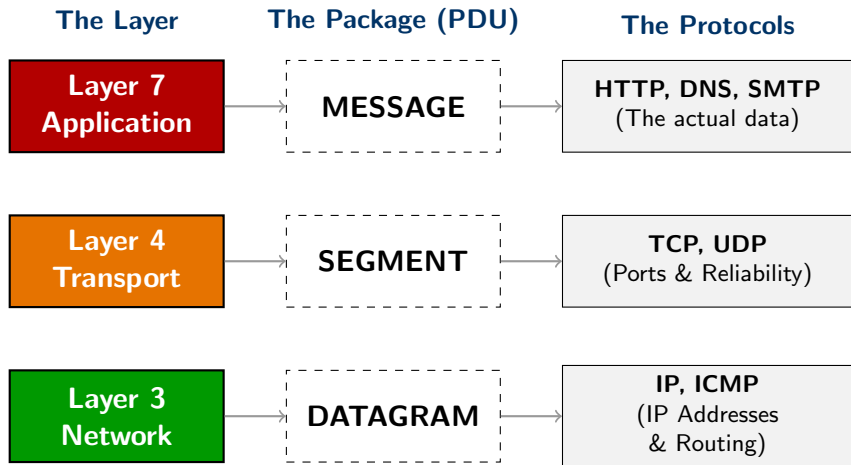
```
... Flags [S], cksum 0x6653 (correct), seq ...
```

What is the Checksum?

- 0x means it is a Hexadecimal number.
- It is a "Digital Verification Seal."
- The Sender does a math calculation on all the bits ($A + B + C = Sum$).
- tcpdump does the exact same math. Because it says **(correct)**, we know the data was not corrupted by electrical noise during travel!

The OSI Map: Layers, Packages, and Protocols

Here is the "Cheat Sheet" mapping the Layer to its Package (PDU) and its Protocols.



Good Luck on the Midterm!